Module 4: Thermal response after radiation energy deposition

Goal: convert the thermalized part of radiation energy deposition into a transient temperature/energy map while retaining finite carrier flight and nonlocal boundary memory.

Macroscopic energy balance

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \boldsymbol{q} = \dot{q}_{\mathrm{th}}}$$

$$\boxed{u \simeq CT}, \qquad \boxed{C\frac{\partial T}{\partial t} + \nabla \cdot \boldsymbol{q} = \dot{q}_{\mathrm{th}}}$$

Here C is volumetric heat capacity and $\dot{q}_{\mathrm{th}}$ is the energy that has actually thermalized into heat carriers.

$$\boxed{\dot{q}_{\mathrm{th}}(x,y,t) = \eta_{\mathrm{th}}\, \dot{q}^{2D}_{\mathrm{rad}}(x,y,t) + Q_{\mathrm{device}}}$$

η_{th} is the thermalization fraction; Q_{device} may include Joule heating or recombination heating.

Why not use Fourier everywhere?

$$\boxed{\boldsymbol{q} = -k\nabla T} \quad \Rightarrow \quad \boxed{C\frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + \dot{q}_{\mathrm{th}}}$$

Fourier conduction assumes local equilibrium and many carrier collisions over the length scale of interest.

$$\boxed{\mathrm{Kn} = \frac{\lambda}{L}}$$

- $\mathrm{Kn} \ll 1$: diffusive, Fourier limit.
- $\mathrm{Kn} \sim 0.1 - 1$: quasi-ballistic, finite-flight and boundary memory matter.
- $\mathrm{Kn} \gtrsim 1$: ballistic carriers cross device-scale distances before scattering.

Module handoff: Module 1 provides $\dot{q}^{2D}_{\mathrm{rad}}$; Module 4 predicts $T(x,y,t)$ and thermal gradients that feed defect kinetics, mobility, recombination, and Module 6 coupling.

Module 4: Microscopic carrier picture and BTE starting point

Heat is carried by microscopic excitations. In silicon, phonons dominate lattice heat transport; energetic electrons/holes may also carry energy before thermalization.

Boltzmann transport equation

$$\boxed{\frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla_{\boldsymbol{r}} f = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} + S_f}$$

In the relaxation-time approximation,

$$\boxed{\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = -\frac{f - f_0}{\tau}}$$

where f is the carrier distribution, f_0 is local equilibrium, and τ is the scattering time.

$$\boxed{\lambda = v\tau}$$

The mean free path λ sets the distance over which the carrier retains memory of its origin.

Ballistic–diffusive split

$$\boxed{f = f_b + f_m}$$

- f_b: boundary-originating or unscattered ballistic component.
- f_m: medium/scattered component that relaxes toward local equilibrium.

For a carrier traveling along ray coordinate s, the ballistic part has the schematic characteristic solution

$$\boxed{f_b(s,t) \sim f_b(s_0, t - \Delta t)\, \exp\left[-\frac{s - s_0}{\lambda}\right]}$$

This expresses two physical effects: finite travel time Δt and attenuation by scattering over distance $s - s_0$.

Physical interpretation: temperature is not purely local immediately after radiation deposition. Some carriers remember where they were emitted and where they scattered before joining the diffusive thermal bath.

Module 4: Reduced ballistic–diffusive thermal equations

The reduced model keeps the computational structure of a heat equation but adds nonlocal forcing from ballistic carriers.

Energy and flux decomposition

$$\boxed{u = u_b + u_m}, \qquad \boxed{\boldsymbol{q} = \boldsymbol{q}_b + \boldsymbol{q}_m}$$

$u_b, \boldsymbol{q}_b$ represent ballistic carrier energy and flux.
$u_m, \boldsymbol{q}_m$ represent the scattered medium part.

The total energy balance becomes

$$\boxed{\frac{\partial(u_b + u_m)}{\partial t} + \nabla \cdot (\boldsymbol{q}_b + \boldsymbol{q}_m) = \dot{q}_{\mathrm{th}}}$$

The medium/scattered part is closed diffusively:

$$\boxed{\boldsymbol{q}_m = -k\nabla T_m}, \qquad \boxed{u_m = CT_m}$$

Effective PDE solved on the 2D domain

$$\boxed{C\frac{\partial T_m}{\partial t} = \nabla \cdot (k\nabla T_m) - \nabla \cdot \boldsymbol{q}_b + \dot{q}_{\mathrm{th}} - \frac{\partial u_b}{\partial t}}$$

The extra terms are the ballistic correction:

- $-\nabla \cdot \boldsymbol{q}_b$: nonlocal carrier energy entering/leaving a point.
- $-\partial u_b/\partial t$: energy stored temporarily in ballistic carriers.
- $\dot{q}_{\mathrm{th}}$: local thermalized energy input from radiation/device physics.

A measured equivalent temperature may be approximated by

$$\boxed{T_{\mathrm{eq}} \simeq \frac{u_b + u_m}{C}}$$

Limit check: if $u_b \to 0$ and $\boldsymbol{q}_b \to 0$, Module 4 reduces to the ordinary 2D heat equation.

Module 4: FEM form, coupling, and project outputs

For the project implementation, Module 4 is a 2D thermal solve driven by radiation deposition and corrected by ballistic/nonlocal energy transport.

Weak form for the medium temperature

$$\boxed{\int_\Omega C\frac{\partial T_m}{\partial t} w \, \mathrm{d}\Omega + \int_\Omega k\nabla T_m \cdot \nabla w \, \mathrm{d}\Omega = \int_\Omega S_{\mathrm{eff}} w \, \mathrm{d}\Omega + \int_{\Gamma_q} \bar{q}\, w \, \mathrm{d}\Gamma}$$

$$\boxed{S_{\mathrm{eff}} = \dot{q}_{\mathrm{th}} - \nabla \cdot \boldsymbol{q}_b - \frac{\partial u_b}{\partial t}}$$

w is a test function, Ω is the 2D device domain, and Γ_q is a heat-flux boundary.

Couplings to the rest of RadiationEffects_proj2

Module 1: $\dot{q}_{\mathrm{rad}}^{2D} \to \dot{q}_{\mathrm{th}}$,
Module 3: $T \to D_j(T),\ k_{\mathrm{ann},j}(T)$,
Module 5: $T \to \mu_n(T), \mu_p(T), R_{\mathrm{SRH}}(T)$,
Module 6: T, ϕ, C_j, n, p solved self-consistently.

Outputs

$\boxed{T(x,y,t)}$, $\boxed{\nabla T(x,y,t)}$, $\boxed{S_{\mathrm{eff}}(x,y,t)}$

Core message: Module 4 replaces a purely local heat-smearing picture with a thermal model that can retain finite carrier flight, scattering, and boundary memory while remaining compatible with FEM-based multiphysics integration.